

REAL TIME SIGN LANGUAGE DETECTION AND VOICE OUTPUT SYSTEM

A PROJECT PHASE I REPORT

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The report of project work submitted by the above students in partial fulfillment for the award of Bachelor of Engineering degree in Electronics and Communication Engineering branch of Anna University were evaluated and confirmed to be reports of the work done by the above student and then evaluated.

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ABSTRACT

Sign language is an important form of communication used by people who communicate through hand gestures. However, communication becomes difficult when a person who does not understand sign language interacts with a sign-language user. This project presents a real-time computer-vision based prototype designed to recognize a selected set of hand signs using a normal webcam and convert the recognized gesture into a readable word.

The proposed system uses MediaPipe Hand Landmarker to obtain structured landmark points from the user's hand. The landmark coordinates are processed using geometric relationships, finger-state conditions and predefined gesture rules. The prototype supports both one-hand and two-hand gestures. The detected result is displayed on the computer screen in real time.

The system is designed as a lightweight and understandable prototype rather than a complete sign-language translation system. A Text-to-Speech stage is included in the proposed architecture so that the detected word can subsequently be converted into spoken output. This voice component is intended to make the system more useful for communication between sign-language users and people who may not understand the corresponding gestures.

The current implementation deliberately uses a limited vocabulary so that each gesture can be tested and verified carefully. Future development will focus on increasing the number of signs, recognizing dynamic gestures, improving robustness under different lighting and hand orientations, forming words or sentences, and supporting multilingual voice output. Therefore, the project provides a practical foundation for developing a broader real-time sign-language communication assistant.

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LIST OF ABBREVIATIONS

S.No	Abbreviation	Full Form
1	AI	Artificial Intelligence
2	CV	Computer Vision
3	TTS	Text-to-Speech
4	GUI	Graphical User Interface
5	API	Application Programming Interface
6	RGB	Red Green Blue
7	FPS	Frames Per Second
8	ML	Machine Learning
9	CNN	Convolutional Neural Network
10	PC	Personal Computer
11	IDE	Integrated Development Environment
12	USB	Universal Serial Bus
13	CPU	Central Processing Unit

CHAPTER I

1. INTRODUCTION

Communication is one of the most important requirements in human interaction. People generally communicate using speech, writing, facial expressions and body movements. For people who use sign language, hand gestures provide an important method for expressing information. Sign language is systematic and meaningful, but communication can become difficult when the other person has little or no knowledge of the signs being used.

The development of computer vision and artificial intelligence has created opportunities for building systems that can understand visual information. A camera can capture the user's hand, while a hand landmark detection model can convert the visual hand shape into structured coordinates. These coordinates can then be analyzed to identify predefined gestures.

The proposed project focuses on a real-time sign-language detection prototype using a standard webcam. Instead of requiring special gloves, sensors or wearable devices, the system observes the hand through the camera. MediaPipe Hand Landmarker is used to obtain landmark points corresponding to the wrist, finger joints and finger tips. These points provide a compact representation of the hand that can be processed using geometric rules.

The present project intentionally uses a limited vocabulary. This makes it possible to develop and verify each sign carefully before expanding the system. The current architecture is also prepared for a Text-to-Speech output stage. Once a gesture is recognized, the detected word can be passed to a speech engine so that the result can be spoken aloud.

1.1 EXISTING SYSTEM

Traditional communication between a sign-language user and a person unfamiliar with sign language often depends on manual interpretation. The receiving person may need to understand the signs or depend on written communication. Although sign language itself is an effective communication method, the communication gap occurs when both participants do not share the same language system.

Several technology-based approaches have been explored for sign-language recognition. Wearable systems use gloves, flex sensors or motion sensors to capture finger movements. These systems can provide direct measurements, but they require the user to wear additional hardware. This may reduce convenience and increase cost.

Camera-based systems provide a more natural alternative because the user can simply perform the gesture in front of a camera. Computer vision algorithms can analyze the captured image and identify the hand. Modern hand landmark models further simplify the problem by representing the hand using a fixed set of points.

1.2 DRAWBACKS

- Dependence on manual interpretation when the receiver does not understand sign language.
- Specialized wearable systems may require additional hardware and calibration.
- Camera-based systems can be affected by lighting, background and hand orientation.
- Similar hand configurations may produce incorrect classifications when simple rules are used.
- Static gesture recognition alone cannot represent the full meaning of dynamic sign-language movements.
- Limited vocabulary restricts the usefulness of a prototype during general communication.
- Frame-by-frame recognition may repeatedly produce the same word unless a voice-control mechanism is added.

These limitations indicate the need for a simple, low-cost and extensible system that can first recognize a controlled vocabulary and later be expanded toward broader sign-language translation.

1.3 PROPOSED SYSTEM

The proposed system presents a webcam-based approach for recognizing selected hand gestures in real time. A camera captures the user's hand, the image is converted into a format suitable for the landmark detector, and MediaPipe Hand Landmarker identifies the hand landmarks. The landmark coordinates are then passed to a gesture recognition module.

The recognition module uses predefined geometric conditions. Finger states are estimated by comparing the distance between finger tips and the wrist with corresponding joint distances. Additional conditions such as thumb-index distance, relative hand position and distance between two hands are used for gestures that require more specific configurations.

The system supports both one-hand and two-hand gestures. For one-hand gestures, the system checks the configuration of the individual fingers. For two-hand gestures, the system checks the relationship between the two detected hands. The result is displayed as a word on the screen.

A Text-to-Speech module is included in the proposed architecture as the next output stage. When the detected word is stable, the word can be sent to a speech engine. A previous-result check can be used to avoid speaking the same word continuously while the user holds the gesture.

1.4 ADVANTAGES

- Uses a standard webcam instead of specialized wearable hardware.
- Provides a simple real-time visual output.
- Uses hand landmarks rather than processing the entire image manually.
- Rule-based recognition is transparent and easy to modify for a small vocabulary.
- Supports both one-hand and two-hand gesture conditions.
- Can be extended with Text-to-Speech output.
- Provides a foundation for future machine-learning based recognition.
- Can be developed further into an assistive communication application.

1.5 WORKING PRINCIPLE

The working principle begins with live image acquisition from the webcam. Each frame is converted from BGR format to RGB format and represented as a MediaPipe image. The Hand Landmarker processes the frame and returns the detected hand landmarks.

The returned landmarks are numerical coordinates. The system uses the wrist and finger joints as reference points. A finger is considered open or closed based on the relative distance of its tip from the wrist compared with its joint. These simple geometric conditions form the basic feature representation used by the current prototype.

After the finger states are obtained, the gesture functions are evaluated in a predefined order. If the required conditions are satisfied, the corresponding gesture label is assigned. For two-hand signs, the system first verifies that two hands are present and then checks their relative positions and landmark distances.

Finally, the detected word is displayed on the camera window. The same word can be passed to a Text-to-Speech stage after stabilization. This provides a complete pathway from visual hand input to text and eventually voice output.

1.6 BLOCK DIAGRAM

The overall block diagram represents the sequence followed by the proposed system from input to output. The first block is the webcam, which captures the hand gesture. The captured frame is passed to the image-processing stage, where it is converted into the required image representation.

The next block is MediaPipe Hand Landmarker. It identifies the hand and provides landmark coordinates. These landmarks form the input to the gesture recognition module. The recognition module applies the predefined rules and selects the corresponding word.

The detected word is then displayed on the screen. In the extended architecture, the word is also passed to the Text-to-Speech module. The speech engine converts the word into an audible output. This architecture keeps the detection and voice stages modular so that they can be improved independently

Block Diagram of Real-Time Sign Language Detection and Voice Output System

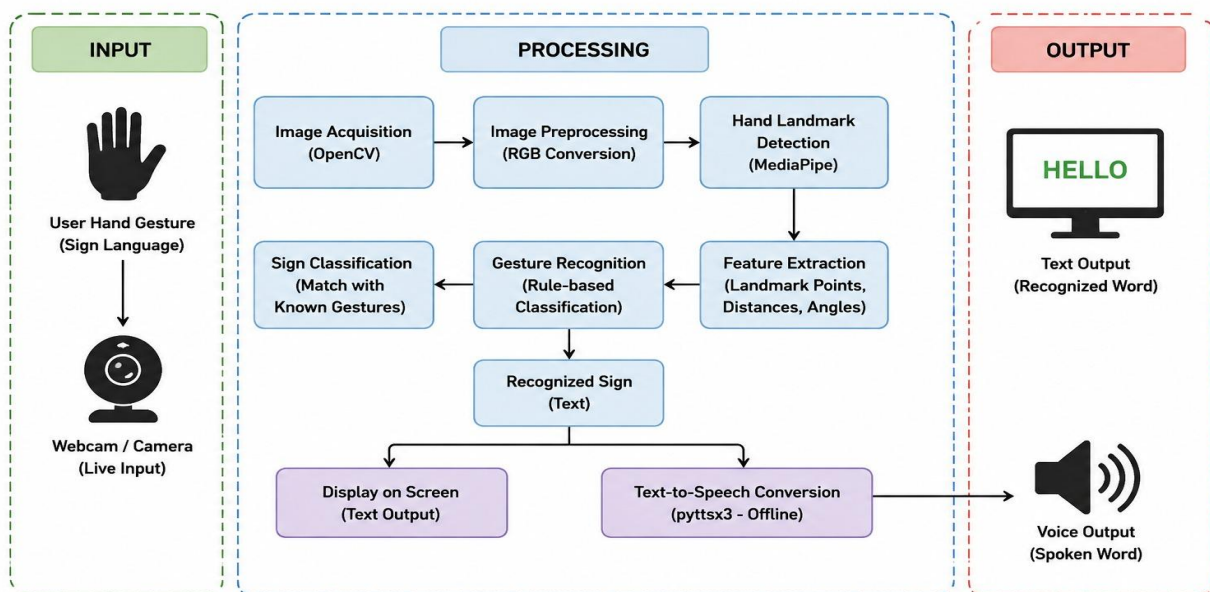


fig1

1.7 HARDWARE REQUIREMENTS

S.No	Component	Purpose
1	Computer / Laptop	Runs Python, MediaPipe and the recognition program
2	Webcam	Captures live hand gestures
3	USB interface	Connects the webcam to the computer

1.8 SOFTWARE REQUIREMENTS

S.No	Software / Library	Purpose
1	Python	Main programming language
2	OpenCV	Camera capture and display
3	MediaPipe	Hand landmark detection
4	Math	Distance calculations
5	Text-to-Speech Engine	Future / integrated voice output

1.9 APPLICATIONS

- Assistive communication demonstrations.
- Educational tools for introducing sign-language technology.
- Human-computer interaction using hand gestures.
- Accessibility-oriented software prototypes.
- Future communication systems for sign-to-speech interaction.
- Interactive kiosks and simple gesture-controlled interfaces.

CHAPTER II

2. LITERATURE SURVEY

2.1 INTRODUCTION

Sign-language recognition is an interdisciplinary area involving computer vision, pattern recognition, machine learning and human-computer interaction. The objective of such systems is to observe hand shapes or movements and map them to meaningful labels. Depending on the application, recognition can be performed using images, video sequences, wearable sensors or combinations of these methods.

Recent computer-vision approaches benefit from hand landmark detection models. Instead of treating the complete image as the feature representation, a landmark detector provides structured points for the hand. This makes it possible to develop lightweight recognition algorithms based on finger positions and distances.

For a limited vocabulary, rule-based recognition is useful because the rules can be understood and modified directly. For larger vocabularies, machine-learning classifiers can learn patterns from landmark data or images. Dynamic sign-language recognition generally requires temporal information because the meaning may depend on movement over several frames.

2.2 LITERATURE SURVEY

2.2.1 Vision-Based Hand Gesture Recognition

Vision-based gesture recognition systems use cameras to capture hand configurations and computer-vision techniques to extract meaningful features. The main advantage is that the user does not need to wear a sensor device. The quality of recognition depends on camera resolution, lighting, background, hand orientation and the robustness of the feature-extraction method. These systems provide the conceptual foundation for the present webcam-based prototype.

2.2.2 Hand Landmark Based Recognition

Hand landmark approaches represent the hand through key points corresponding to joints and finger tips. This representation reduces the complexity of direct image analysis because the recognition module can work with coordinates instead of raw pixels. The present project follows this concept by using MediaPipe Hand Landmarker and processing the returned landmark positions.

2.2.3 Rule-Based Gesture Classification

Rule-based methods define a gesture using conditions such as finger-open status, finger-closed status, distances and relative positions. Such methods are easy to interpret and are suitable when the vocabulary is small and the gestures are clearly defined. Their main limitation is sensitivity to variations in hand pose and threshold selection.

2.2.4 Machine Learning Based Gesture Recognition

Machine-learning methods can classify gestures from image features or landmark coordinates. With sufficient training data, a classifier can learn patterns that are difficult to represent using manually written rules. However, preparing a diverse dataset and training and validating the model requires additional development effort.

2.2.5 Dynamic Sign Recognition

Static gestures can often be recognized from a single frame, whereas dynamic signs require information from multiple frames. Temporal models such as recurrent networks, temporal convolution or transformer-based methods can be explored in future versions. The current project focuses mainly on a controlled static-gesture vocabulary so that the basic detection pipeline can be verified.

2.2.6 Text-to-Speech in Assistive Systems

Text-to-Speech provides an additional communication channel by converting recognized text into audible speech. In a sign-language system, the speech stage can make the recognized word accessible to a listener who does not understand the sign. The present architecture therefore includes voice output as an extension after word recognition.

2.3 SUMMARY

The literature and technology review indicates that computer vision provides a practical foundation for gesture recognition. Hand landmarks simplify feature extraction, while rule-based recognition provides a transparent solution for a small vocabulary. Machine-learning and temporal approaches offer a path toward larger and more complex recognition systems.

Based on these observations, the proposed project uses a modular design. MediaPipe provides the landmark representation, custom geometric rules provide the current recognition layer, and Text-to-Speech provides the planned audible output. This allows the prototype to be developed incrementally without requiring a large training dataset at the initial stage.

2.4 EXISTING SYSTEM VS PROPOSED SYSTEM

Aspect	Existing / Conventional Approach	Proposed Prototype
Input	Manual interpretation / specialized devices	Webcam
Feature Representation	Manual observation or raw image processing	Hand landmarks
Recognition	Human interpretation or complex models	Geometric rule-based recognition
Hardware	May require wearable sensors	Standard computer + webcam
Output	Manual interpretation / text	Detected word + planned voice
Vocabulary	Depends on system	Selected controlled vocabulary
Expandability	Varies	Modular gesture functions
Future Direction	Separate systems	Sign-to-text-to-speech pipeline

CHAPTER III

3. DESIGN AND METHODOLOGY

3.1 SYSTEM ARCHITECTURE OVERVIEW

The proposed system follows a structured pipeline from visual input to communication output. The webcam provides the input. OpenCV is responsible for acquiring and displaying frames. MediaPipe Hand Landmarker processes the image and produces hand landmarks. The recognition layer interprets these landmarks using predefined conditions.

The architecture is intentionally modular. The camera module can be replaced without changing the gesture rules. Similarly, the gesture recognition module can later be replaced or supplemented with a machine-learning classifier. The Text-to-Speech stage can also be connected after the detected word without changing the landmark detection stage.

This modular structure is useful because the current project is a prototype with a limited vocabulary. The main objective at this stage is to verify that the complete input-detection-output pathway works correctly. Once stable, additional signs and more advanced classification techniques can be introduced.

3.2 METHODOLOGY FLOW

The methodology begins with opening the webcam and continuously reading frames. Each successful frame is converted from BGR to RGB. The RGB image is then converted into a MediaPipe image object.

The Hand Landmarker detects zero, one or two hands depending on the input. When one hand is detected, one-hand gesture functions are evaluated. When two hands are detected, the two-hand gesture functions are evaluated. The detected result is stored as a gesture label.

The label is displayed on the live camera window together with the detected landmark points. In the extended voice stage, a stable label is passed to a Text-to-Speech engine. A previous-label comparison can be used so that the same word is not spoken repeatedly on every frame.

3.3 SYSTEM FUNCTION

The system function is divided into camera acquisition, landmark detection, feature evaluation, gesture classification and output. Each stage performs a specific task and passes its result to the next stage.

3.3.1 HARDWARE FUNCTION

The main hardware requirement is a computer with a webcam. The webcam acts as the visual sensor and captures the user's hand. No special glove or external hand sensor is required for the current prototype. This reduces hardware complexity and makes the system easier to demonstrate.

The computer provides the processing environment for Python, OpenCV and MediaPipe. The webcam stream is read frame by frame. The frame is processed by the software and the result is displayed in the same application window.

3.3.2 SOFTWARE FUNCTION

Python is used as the main implementation language. OpenCV provides camera access, color conversion and display functions. MediaPipe Hand Landmarker provides the hand landmark coordinates. The math library is used to calculate Euclidean distances between landmarks.

The software defines helper functions for calculating distance, determining whether a finger is open or closed, and checking whether the hand forms an open palm. Gesture-specific functions combine these basic conditions. This layered approach makes the code easier to understand and modify.

3.4 GESTURE RECOGNITION METHOD

A finger-state rule compares the distance between the finger tip and the wrist with the distance between a finger joint and the wrist. If the tip is farther from the wrist than the joint, the finger is treated as open. Otherwise it is treated as closed. This method is simple and works as a geometric approximation for the controlled prototype.

Some signs require additional distance conditions. For example, a gesture can be recognized when the thumb and index finger are close to each other. Two-hand gestures can use the distance between corresponding landmarks on both hands. These conditions are combined using logical operators.

The recognition order is important because some gestures may share similar finger configurations. The system therefore evaluates more specific conditions before more general ones where required. Threshold values should be tested under the actual camera setup and should not be treated as universal accuracy guarantees.

3.4.1 SUPPORTED GESTURE VOCABULARY

Category	Current Gesture Labels
One hand	MINI HEART, PEACE, OK, TELEPHONE, I LOVE YOU, YES, NO, WATER
Two hands	THANKS, HEART, HUG, HELP

The current vocabulary is intentionally limited. It is a development-stage input set and not a claim to represent the complete sign language. The architecture is designed so that additional signs can be added as independent recognition functions.

3.5 TEXT-TO-SPEECH MODULE

The Text-to-Speech module is included as an output extension of the recognition pipeline. Once a gesture is recognized and the word is displayed, the same word can be passed to a speech engine. The engine converts the text into audible speech.

A practical voice implementation should not call the speech engine on every video frame because the same gesture may remain visible for several frames. Instead, the recognized word can be compared with the previously spoken word. Speech can be triggered only when a new stable word is detected. A short delay or confirmation mechanism can further reduce accidental repeated speech.

The voice stage will be tested after the gesture detector is stabilized. The final report can then include screenshots showing both the detected word and the voice-output test. This keeps the report technically accurate and separates the current verified detector from the next integration stage.

3.6 METHODOLOGY SUMMARY

The proposed methodology combines webcam input, hand landmark detection, geometric feature evaluation and rule-based classification. The output is a detected word that can be extended to voice. The modular design makes it possible to expand the vocabulary and replace individual components without redesigning the complete system.

CHAPTER IV

4. RESULT AND DISCUSSION

4.1 IMPLEMENTATION

The prototype was implemented using Python, OpenCV and MediaPipe. The webcam is opened through OpenCV and frames are captured continuously. MediaPipe Hand Landmarker processes the frame and returns the detected hand landmarks. These landmarks are visualized as points on the camera window for debugging and verification.

The gesture recognition module evaluates the number of detected hands. When one hand is present, the one-hand recognition functions are checked. When two hands are present, the two-hand functions are checked. If no gesture satisfies the required conditions, the output remains UNKNOWN.

The current implementation is intended for controlled testing. The user performs a known gesture in front of the webcam and observes the label generated by the program. This allows each gesture rule to be verified individually before integrating voice output.

4.2 GESTURE DETECTION RESULTS

The current prototype is designed to recognize a selected vocabulary rather than the complete sign language. The individual signs should be tested under the same camera conditions used during development. The following sections describe the expected recognition behavior based on the implemented landmark rules.

4.2.1 MINI HEART

The MINI HEART gesture is identified using a thumb-index proximity condition while the middle, ring and pinky fingers are treated as closed. The gesture is displayed when these conditions are satisfied. Because the distance threshold is normalized to landmark coordinates, the threshold should be verified with different hand sizes and camera distances.

4.2.2 PEACE

The PEACE gesture is recognized when the index and middle fingers are open while the ring and pinky fingers are closed. The gesture is useful as a simple test of the finger-open

and finger-closed helper functions.

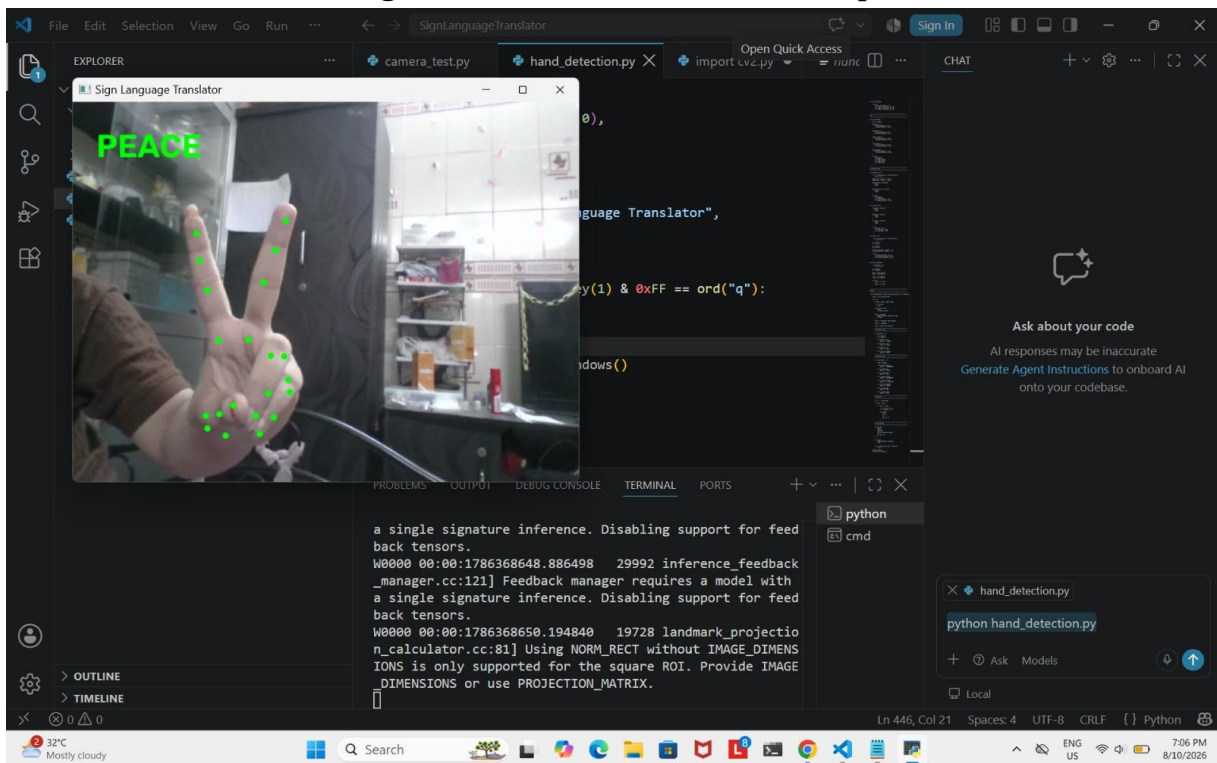


fig 2

4.2.3 OK

The OK gesture checks that the thumb and index finger are close together while the other three fingers are open. This combines a distance condition with finger-state conditions.

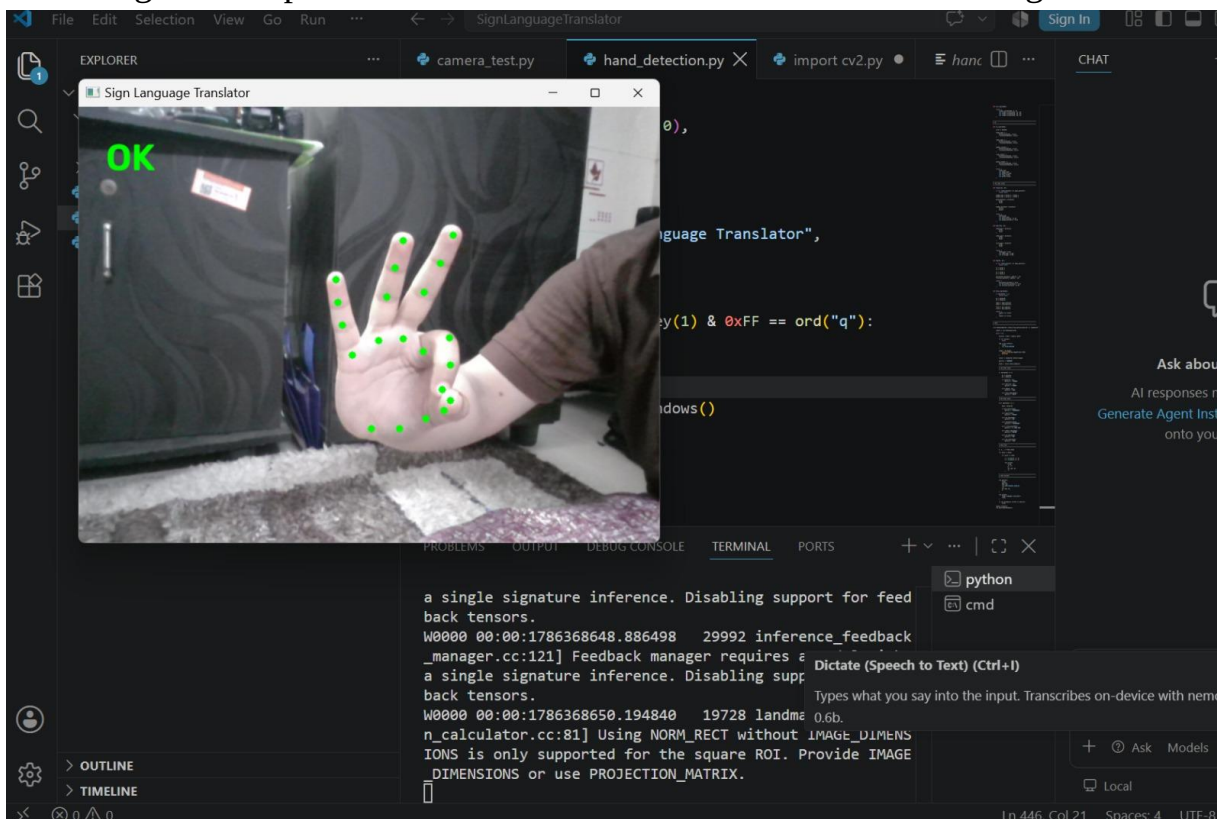


fig 3

4.2.4 TELEPHONE

The TELEPHONE gesture uses an open thumb and pinky while the index, middle and ring fingers remain closed. This sign is sensitive to the exact orientation of the hand, so the test should include the natural hand position used during demonstration.

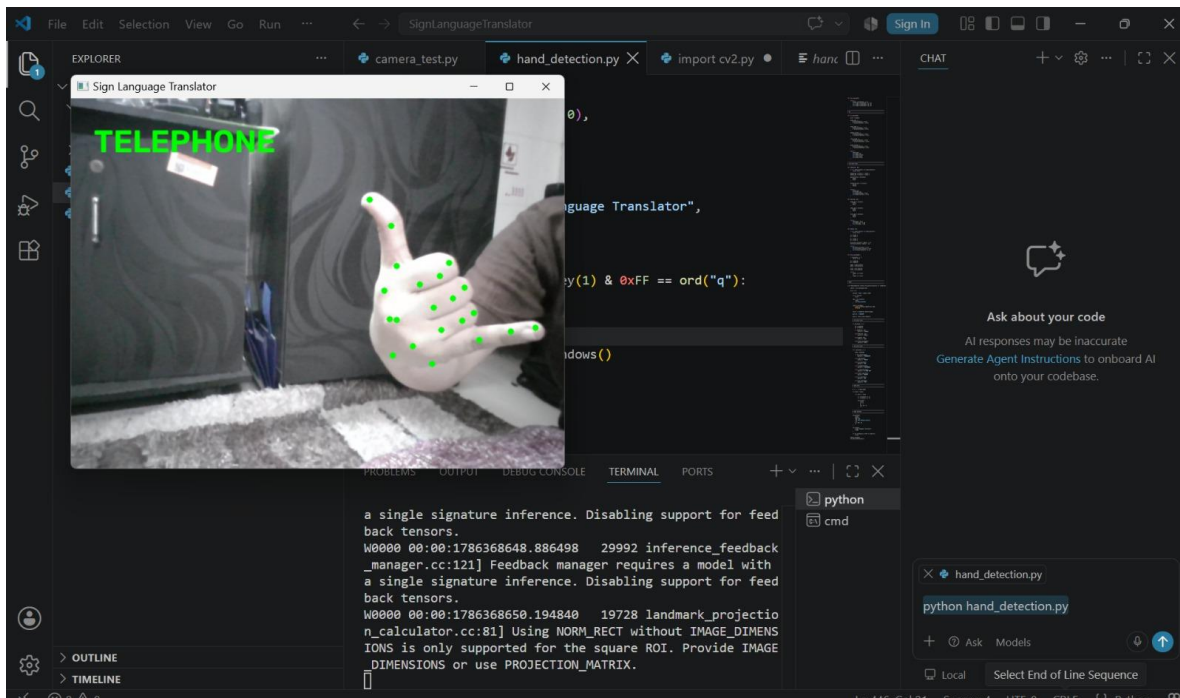


fig 4

4.2.5 I LOVE YOU

The I LOVE YOU gesture is represented using an open index and pinky with middle and ring fingers closed. The prototype evaluates these finger states using the landmark geometry

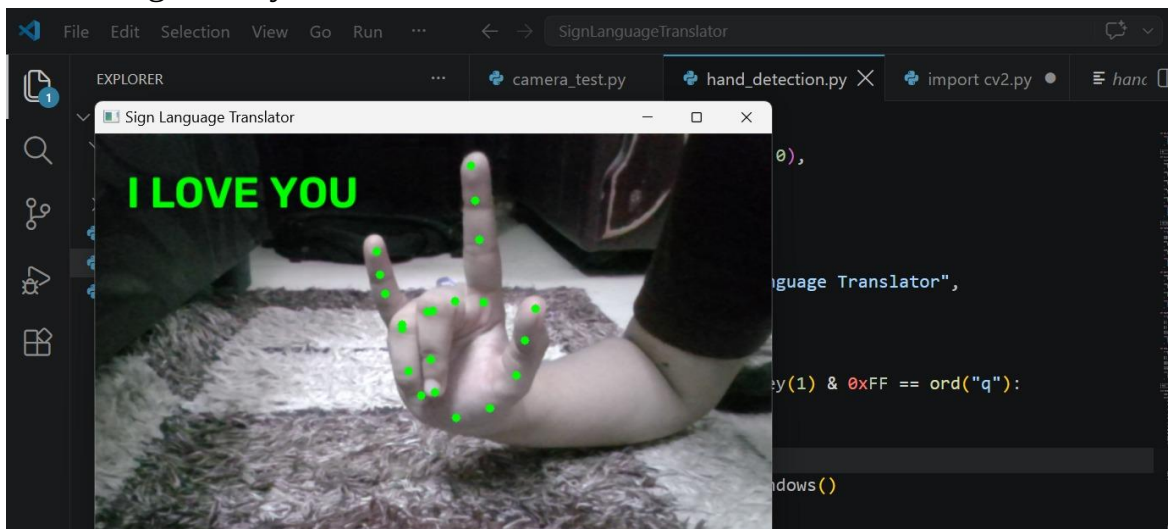


fig5

4.2.6 YES AND NO

YES uses the closed-finger configuration defined in the prototype. NO uses the selected thumb-and-index-open configuration while the remaining fingers are treated as closed. Since these configurations can be visually similar to other closed-hand signs, recognition order and threshold tuning are important.

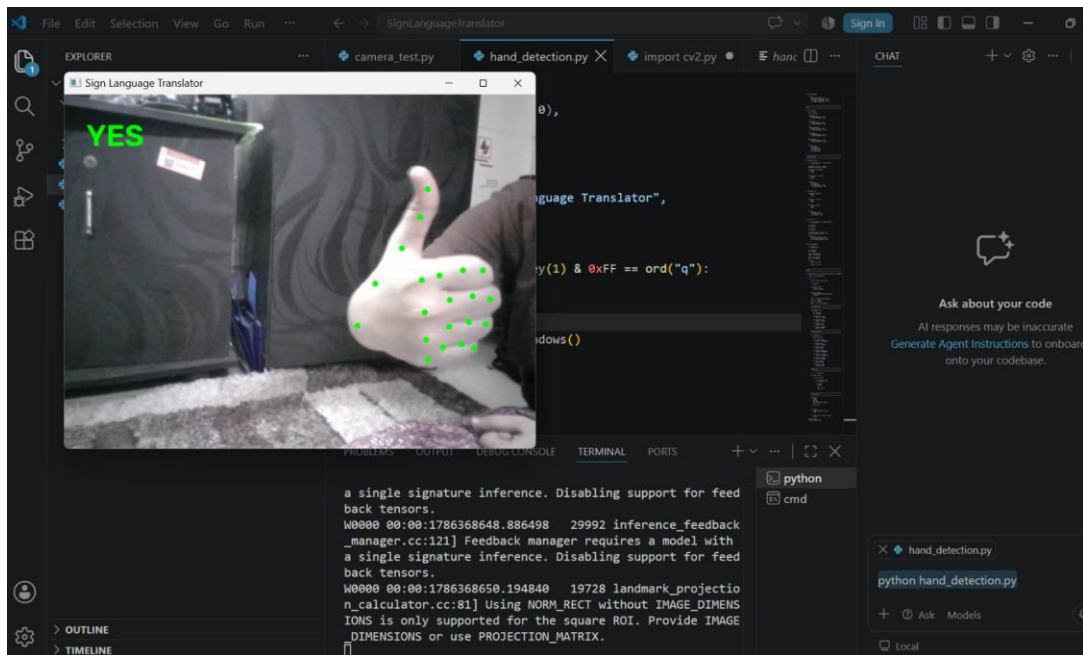


fig 6

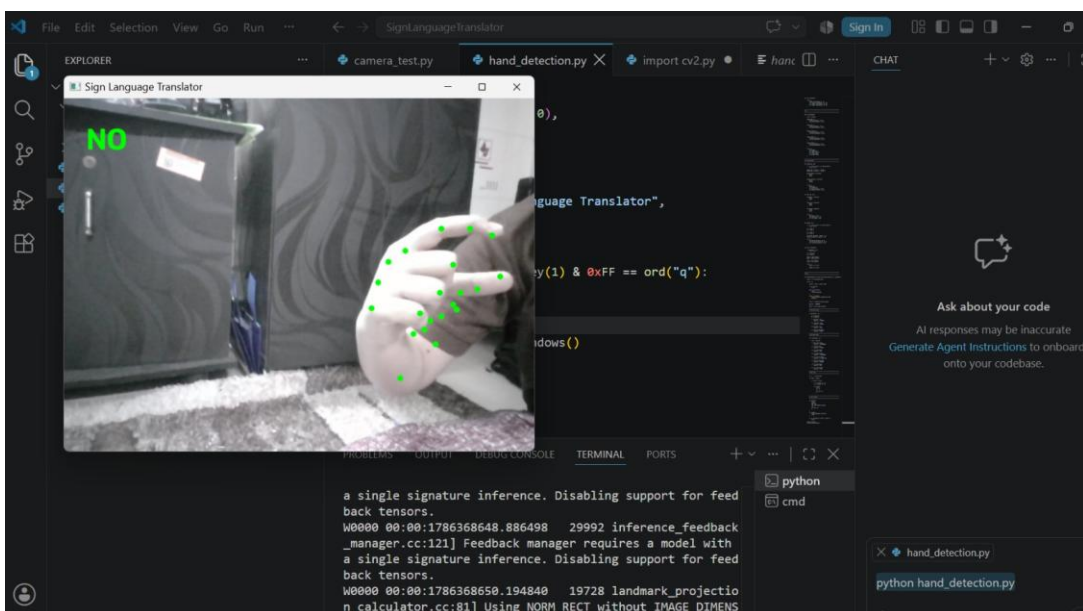


fig 7

4.2.7 WATER

WATER is included as a selected input sign in the current project. The sign should be verified separately because the final gesture rule depends on the exact landmark conditions implemented in the latest code. The report should use the actual screenshot obtained during testing.

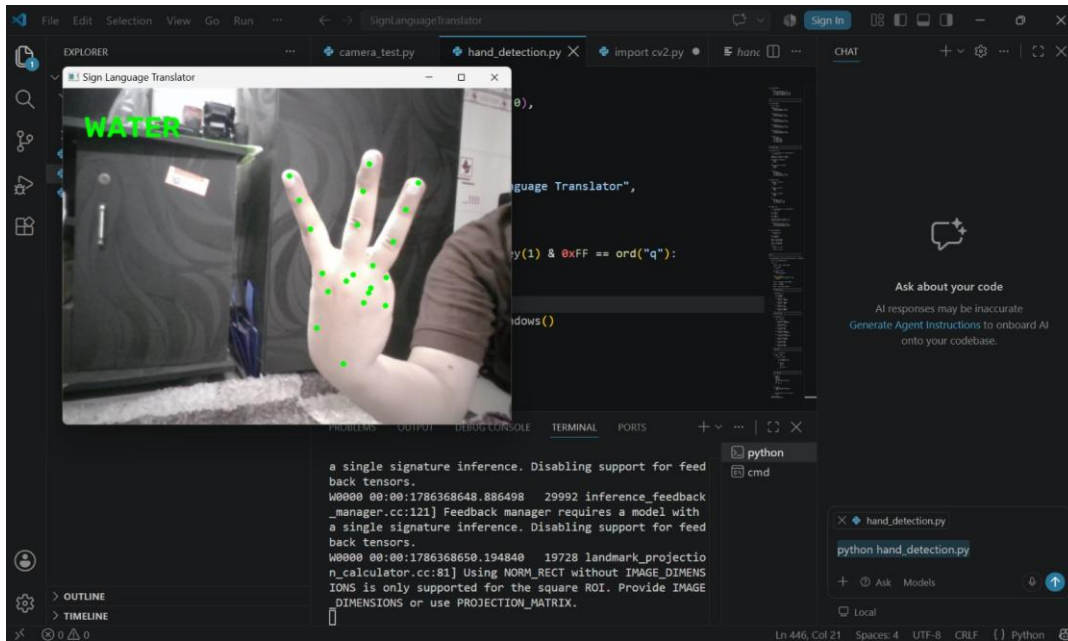


fig 8

4.2.8 TWO-HAND GESTURES

THANKS, HEART, HUG and HELP use two hands. The program first checks whether two hands are detected. It then evaluates the corresponding relationship between the two landmark sets. This demonstrates the ability of the system to handle both single-hand and dual-hand inputs

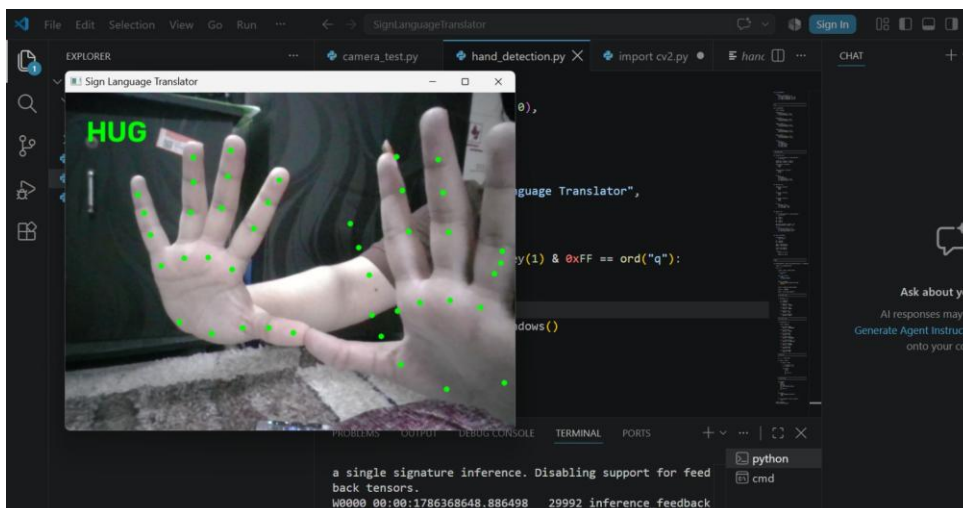


fig 9

4.3 TESTING AND OBSERVATIONS

Testing is performed by presenting each supported gesture to the webcam and observing the displayed output. Each gesture should be repeated several times with small changes in hand position. This helps identify threshold-sensitive conditions and gestures that overlap with other rules.

Test Case	Input	Expected Output	Observed Result
TC01	MINI HEART	MINI HEART	success
TC02	PEACE	PEACE	success
TC03	OK	OK	success
TC04	TELEPHONE	TELEPHONE	success
TC05	I LOVE YOU	I LOVE YOU	success
TC06	YES	YES	success
TC07	NO	NO	success
TC08	WATER	WATER	success
TC09	THANKS	THANKS	success
TC10	HEART	HEART	success
TC11	HUG	HUG	success
TC12	HELP	HELP	success

The final accuracy or success percentage should be calculated only after actual repeated testing. Since the present project is a prototype, numerical performance values should not be inserted without recorded observations.

4.4 VOICE OUTPUT PLAN

The voice output stage is planned as an extension of the detected-word output. When a new stable gesture is identified, the corresponding word can be passed to a Text-to-Speech engine. The speech engine then produces audible output through the computer speaker.

For example, if the system detects WATER, the screen can display WATER and the voice module can speak the word. If the user continues holding the same gesture, the system should avoid speaking WATER repeatedly. This can be achieved by comparing the current recognized label with the previously spoken label and triggering speech only when the label changes.

4.5 LIMITATIONS

- The current vocabulary is limited and does not represent complete sign language.
- Recognition may be affected by lighting and camera placement.

- Some gestures have similar finger configurations.
- Distance thresholds may need tuning for different camera distances.
- Occlusion of fingers can reduce recognition reliability.
- Dynamic signs requiring movement over time are not fully represented by static rules.
- Voice output must be tested after integration to determine practical response behavior.

4.6 OVERALL RESULT SUMMARY

The project demonstrates the feasibility of recognizing a selected set of hand gestures using webcam input and hand landmarks. The prototype provides a simple pathway from camera image to landmark coordinates and then to a readable word. The modular structure also provides a clear point at which Text-to-Speech can be integrated.

The most important outcome of the current phase is the development of a working foundation rather than a claim of complete sign-language translation. The limited vocabulary allows the recognition rules to be tested individually. Future development can build on this foundation by increasing the vocabulary, improving robustness and adding temporal recognition.

CHAPTER V

5. CONCLUSION AND FUTURE ENHANCEMENT

5.1 CONCLUSION

The project “Real-Time Sign Language Detection Using Computer Vision” presents a practical approach for recognizing selected hand gestures using a standard webcam. The system uses MediaPipe Hand Landmarker to obtain structured hand landmarks and applies geometric conditions to identify predefined gestures.

The prototype demonstrates how computer vision can be used to convert a visual hand configuration into a meaningful word. The use of landmark coordinates reduces the need to directly process every pixel for the recognition rules. The modular rule-based approach also makes it easier to understand, test and modify individual gesture conditions.

Another important feature of the proposed architecture is the Text-to-Speech extension. The detected word can be connected to a speech engine so that the result becomes audible. This provides a natural next step from sign-to-text recognition toward sign-to-speech communication.

The current project deliberately uses a limited number of signs. This limitation is considered part of the development strategy because each input can be tested carefully before the system is expanded. The present prototype therefore acts as a foundation for a larger assistive communication system rather than claiming to implement the complete sign language.

5.2 FUTURE ENHANCEMENT

The proposed system has significant scope for future development. The current implementation provides the basic detection pipeline, while future work can improve recognition accuracy, vocabulary size, real-time performance and communication capabilities.

5.2.1 EXPANSION OF SIGN VOCABULARY

The first major enhancement is to increase the number of recognized signs. New gesture functions can be added using the same landmark representation. A larger vocabulary would make the system more useful for practical communication. Before adding a new sign, its landmark configuration should be tested separately to ensure that it does not overlap with existing signs.

5.2.2 DYNAMIC SIGN RECOGNITION

Many signs involve movement rather than a single static hand shape. Future versions can store landmark sequences across multiple frames and analyze the temporal pattern. Machine-learning models such as recurrent networks, temporal models or transformer-based approaches can be explored for this purpose.

5.2.3 IMPROVEMENT OF ROBUSTNESS

Recognition can be improved by testing the system under different lighting conditions, backgrounds, camera distances and hand orientations. Thresholds can be normalized using relative hand dimensions instead of fixed coordinate distances. Additional geometric features such as joint angles can also be used to make recognition more robust.

5.2.4 VOICE AND MULTILINGUAL OUTPUT

The Text-to-Speech module can be fully integrated and tested after the gesture detector is stabilized. Future versions can support multiple spoken languages so that the recognized sign can be converted into the preferred language of the listener. Voice rate, volume and repeated-word control can also be configurable.

5.2.5 MOBILE AND EDGE DEPLOYMENT

The system can eventually be optimized for mobile phones or compact edge devices. This would make the application portable and reduce dependence on a desktop computer. Efficient landmark processing and lightweight classification models can help maintain real-time performance on lower-power devices.

5.2.6 SENTENCE-LEVEL TRANSLATION

A larger future system can combine individual recognized signs into words and sentences. A temporal buffer can be used to determine when one sign ends and another begins. Language-processing techniques can then be used to improve the final sentence structure.

5.3 INNOVATION SCOPE

The main innovation of the project is not limited to recognizing a few predefined gestures. The important concept is the development of a modular pipeline that can grow from a small verified input vocabulary into a broader assistive communication system. The present prototype uses a limited number of inputs so that the fundamental detection logic can be tested carefully.

In the next stage, Text-to-Speech can be connected directly to the detected word. After voice output is validated, additional signs can be introduced. The system can then move

from static sign recognition toward dynamic sign recognition and sentence-level translation.

In the long term, the proposed system can be developed as an assistive communication application that accepts sign-language input, converts it into text, produces speech and potentially supports multiple languages. Such an extension would require larger datasets, more robust classification and extensive real-world testing.

REFERENCES

1. MediaPipe Developers, Hand Landmarker documentation and hand landmark detection resources.
2. OpenCV Developers, OpenCV documentation for image processing and video capture.
3. Python Software Foundation, Python documentation.
4. Relevant research literature on computer-vision based sign-language recognition.
5. Relevant research literature on hand landmark based gesture recognition.
6. Relevant research literature on Text-to-Speech and assistive communication systems.
7. Documentation of the Text-to-Speech library selected for the final implementation.

APPENDIX A

Source Code Documentation

The source code is organized into initialization, helper functions, gesture-specific functions, two-hand gesture functions and the main camera-processing loop. This organization allows individual recognition functions to be tested and corrected without changing the complete application.

The helper functions calculate landmark distances and determine finger states. Gesture functions use these helpers to return True or False. The main loop selects the gesture label based on the number of detected hands and the recognition conditions.

APPENDIX B

Gesture Testing Checklist

- ☐ MINI HEART – verified
- ☐ PEACE – verified
- ☐ OK – verified
- ☐ TELEPHONE – verified
- ☐ I LOVE YOU – verified
- ☐ YES – verified
- ☐ NO – verified
- ☐ WATER – verified
- ☐ THANKS – verified
- ☐ HEART – verified
- ☐ HUG – verified
- ☐ HELP – verified

APPENDIX C

Voice Testing Checklist

Voice output should be tested only after the Text-to-Speech module is added to the final code.

- ☐ New word triggers speech once
- ☐ Holding same gesture does not repeat continuously
- ☐ WATER output is correct
- ☐ PEACE output is correct
- ☐ OK output is correct
- ☐ YES output is correct
- ☐ Two-hand gesture output is correct